

{EPITECH}

<SINCE 1999/>

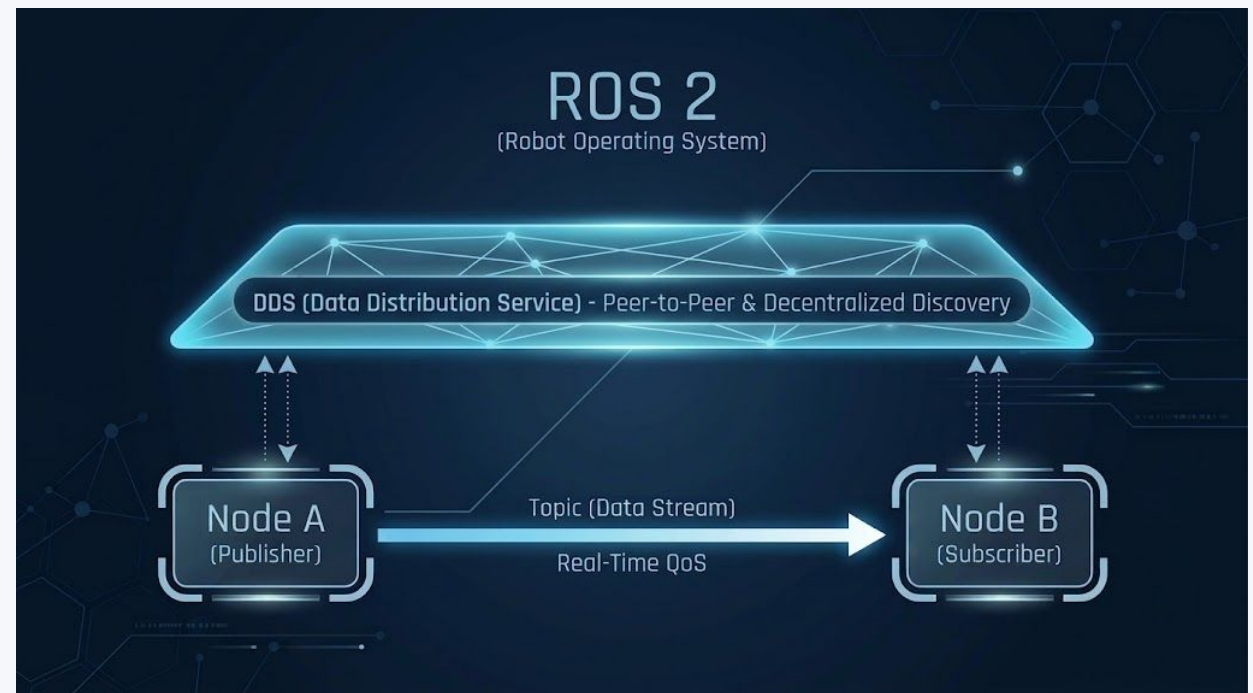
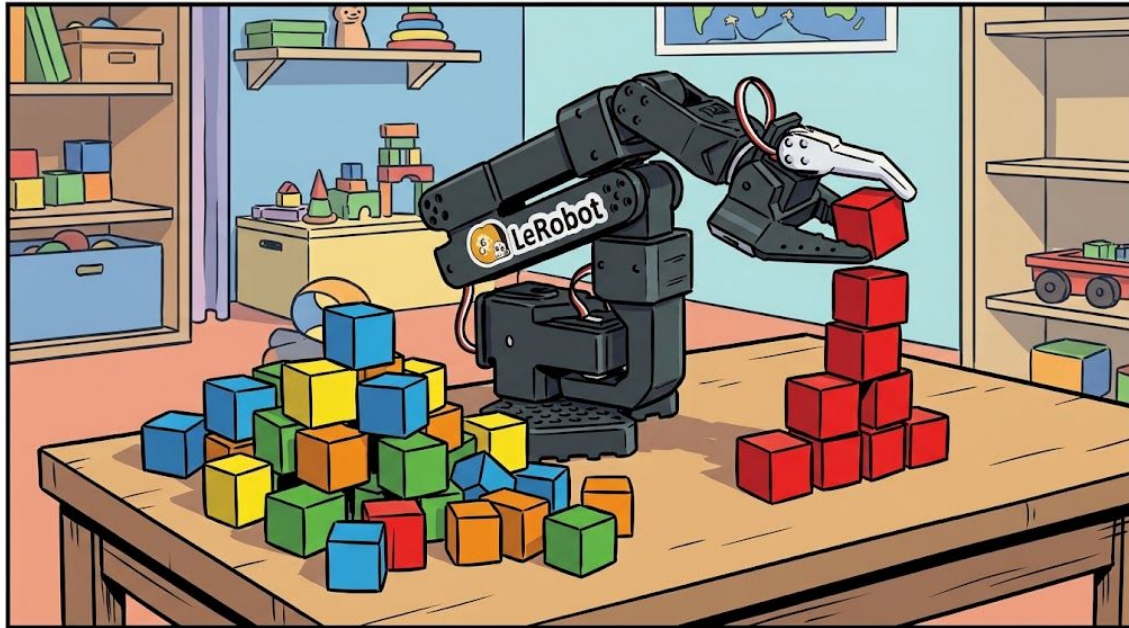
< Robotique TEK5 · Day 2 />

Bootstrap: Yesterday

{ <TECH TOGETHER TOMORROW> }

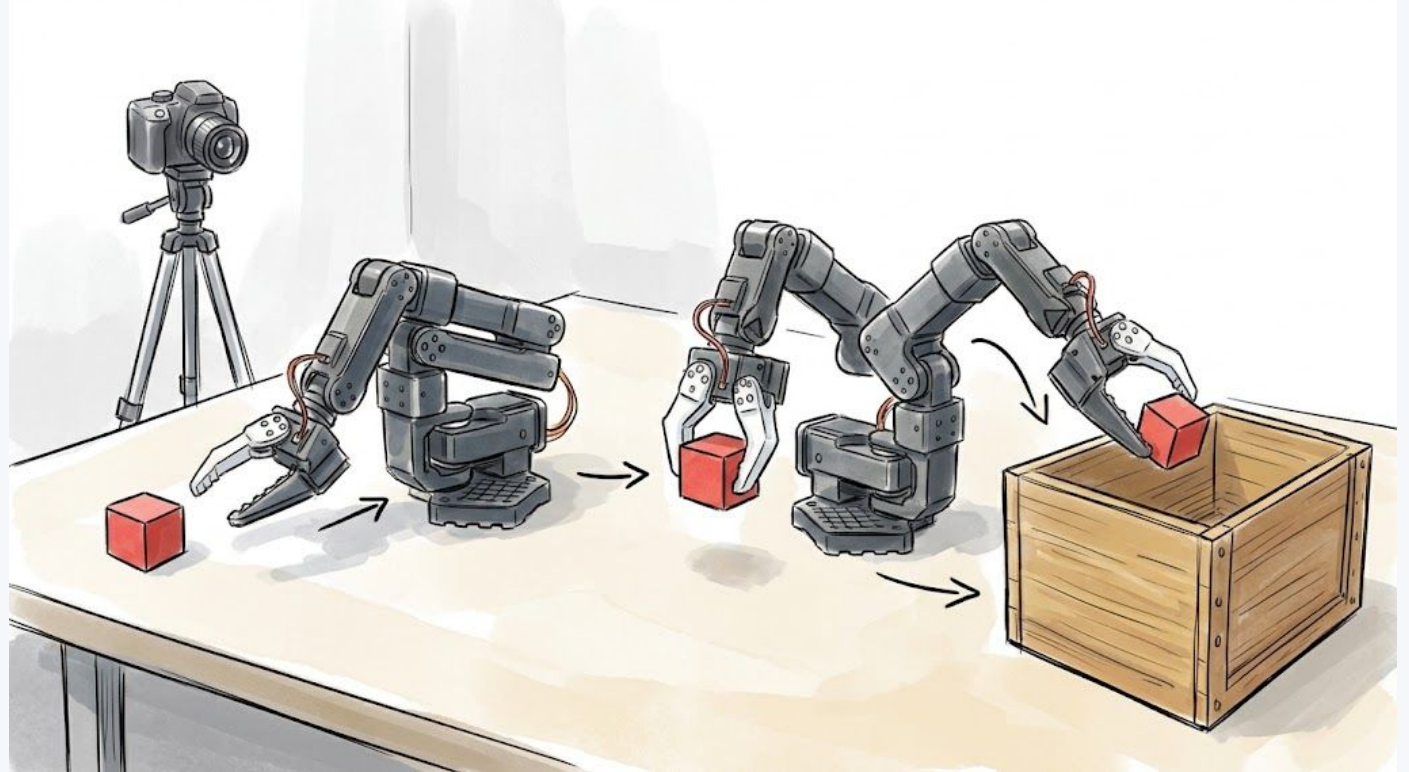


< ROS2 and the robot arm SO101 />



< The goal />

See the object.
Grasp it.
Drop it in the box.



With an external camera, an SO-101 arm and a ROS 2 pipeline you build.

In MuJoCo simulation first, then on the real arm.

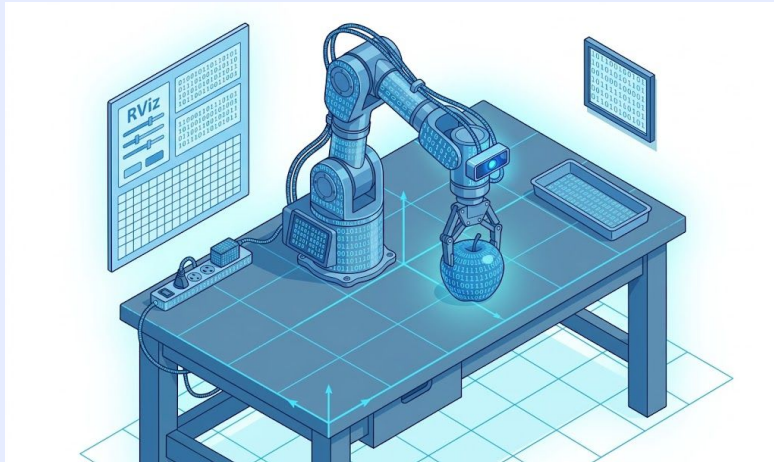
< Move it in Simulation, then make it real />

```
ros2 topic pub --once /joint_command sensor_msgs/msg/JointState '{name: [shoulder_pan], position: [0.5]}'
```

SO101 - Simulation

MuJoCo

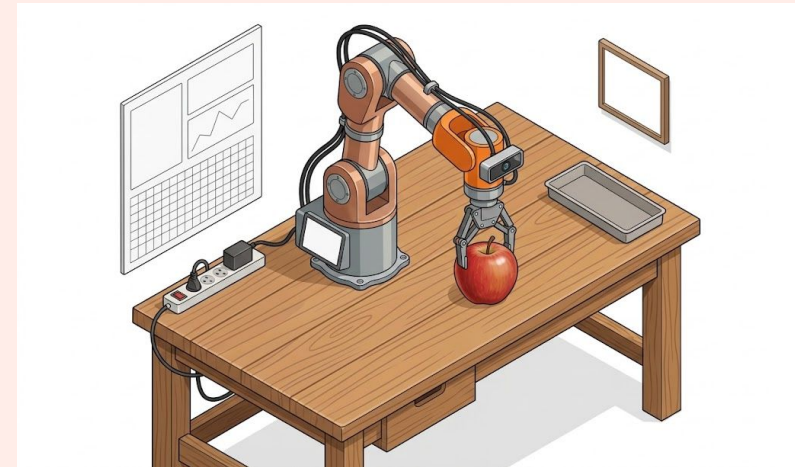
- All your code will be tested here
- Safe



SO101 - Real

The real arm

- Be careful (about the arm but also the screens and laptop around)
- Real validation of your work (and fun part of this project)



Make it easy to switch from simulation to reality in your code

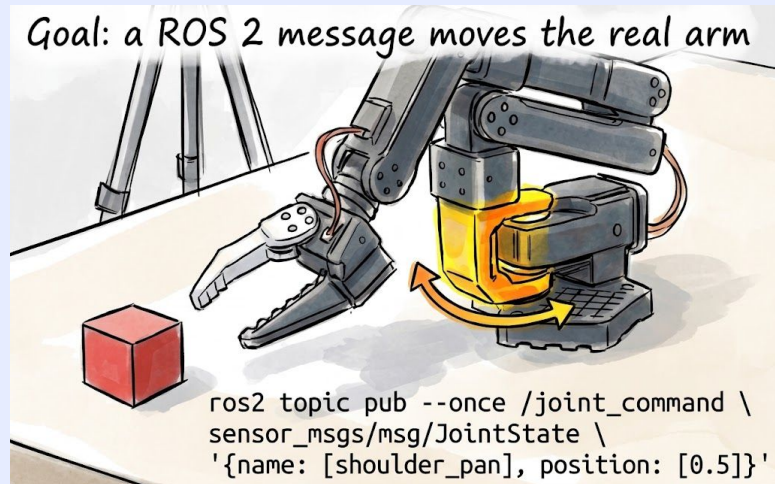
Bootstrap: make it move



MORNING

Make it move

- Setup: Docker, sim, RViz
- Your driver node
- Real arm: connect, calibrate



AFTERNOON

Start the project

- Organize your team (3 per team)
- Explore the sim and IK



We go around the groups all day. Ask questions.

< Let's Dive into the Driver Node />



< Setup />

```
cd docker
docker compose build    # if not done last night
xhost +local:docker
docker compose up -d control perception viz
docker compose logs -f control    # wait: Brain node prêt
docker compose exec control bash

# in every new shell
source /opt/ros/jazzy/setup.bash
source /ros2_ws/install/setup.bash

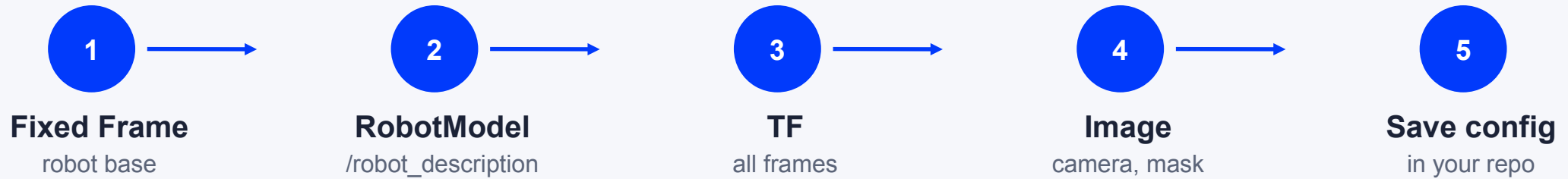
# check
ros2 topic list
ros2 topic hz /joint_states
```

You need

- Docker + Compose
- X11 for RViz
- VM: USB passthrough for the arm

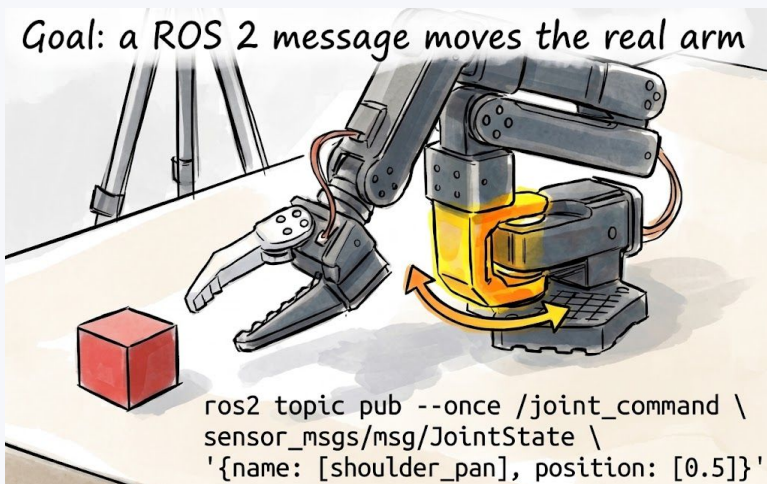
Something missing? Tell us now, not at the end of the morning.

< Sim & RViz />



First move

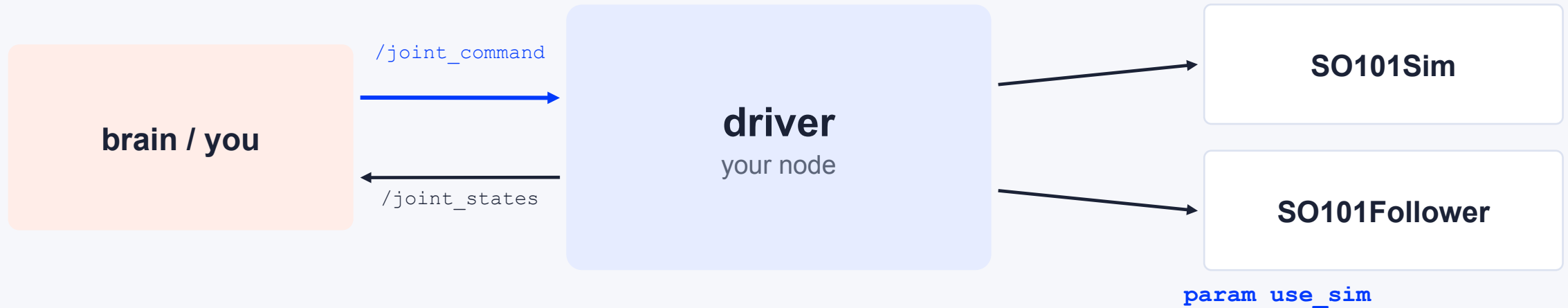
```
ros2 topic pub --once /joint_command \
  sensor_msgs/msg/JointState \
  '{name: [shoulder_pan], position: [0.5]}'
```



Toolbox

```
ros2 topic list
ros2 topic echo <t>
ros2 topic hz <t>
ros2 topic info -v <t>
ros2 node info <n>
ros2 interface show <msg>
rqt_graph
```

< Your driver node />



Definition of done

☐

`/joint_states` at a steady rate

☐

RViz arm follows `/joint_command`

☐

Radians in and out, gripper in %

☐

Partial command: only named joints move

☐

`use_sim` switches the backend only

☐

Joint limits clamped

< The real arm />

Safety first

- Clear the workspace
- Small moves first
- Hands off while it moves
- Know where the power plug is
- Test every new command in sim first

```
# Calibration has been done already

# find the port
ls /dev/ttyACM*

# start the real-arm profile
# -> launches driver with use_sim:=false, then the brain
docker compose --profile real up -d control-real

# check: "Mode : bras réel" then "Bras connecté"
ros2 run <your_pkg> driver --ros-args -p use_sim:=false
```

MORNING CHECKPOINT

Show us your real arm moving from a ROS 2 topic

< Milestones today />

1

Installation

Docker

2

Rviz

Add model
Show the robot

3?

Joint states

Show the joint states / TF
(Simulation)

4?

Real robot

Connect it
Move the robot in live and
see it in rviz

5

Publish joint command

1. Simulation
2. Real

6 Bonus

Forward Kinematics

Calculate end-effector
position XYZ from joint states
(show it on rviz)

7 Bonus

Inverse Kinematics

Move end-effector to XYZ
position.

8

Exploration

your application

< Afternoon: start the project />

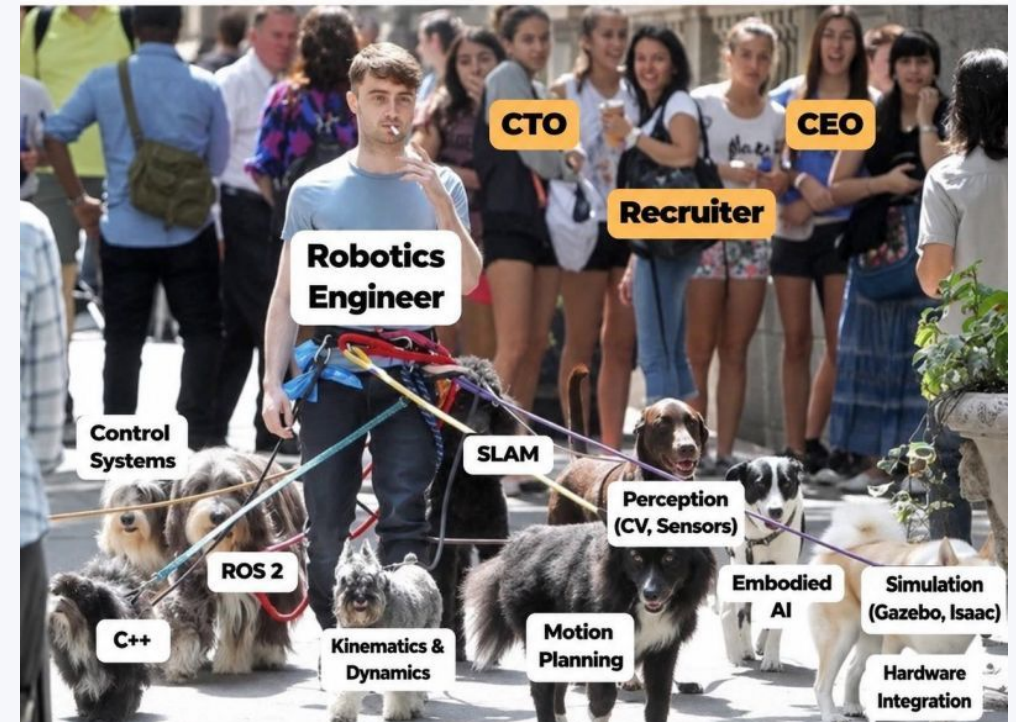
Your arm reaches a target position in sim

Your IK, your choice: ikpy, **Pinocchio**, analytical, your own solver...

Organize your team

- Split roles ?
 - driver → everybody ?
 - sim & IK
 - perception
 - brain & toolingBut every part is closely related.
- One Git repo

Being a Robotics Engineer in 2026....



{EPITECH}

<SINCE 1999/>

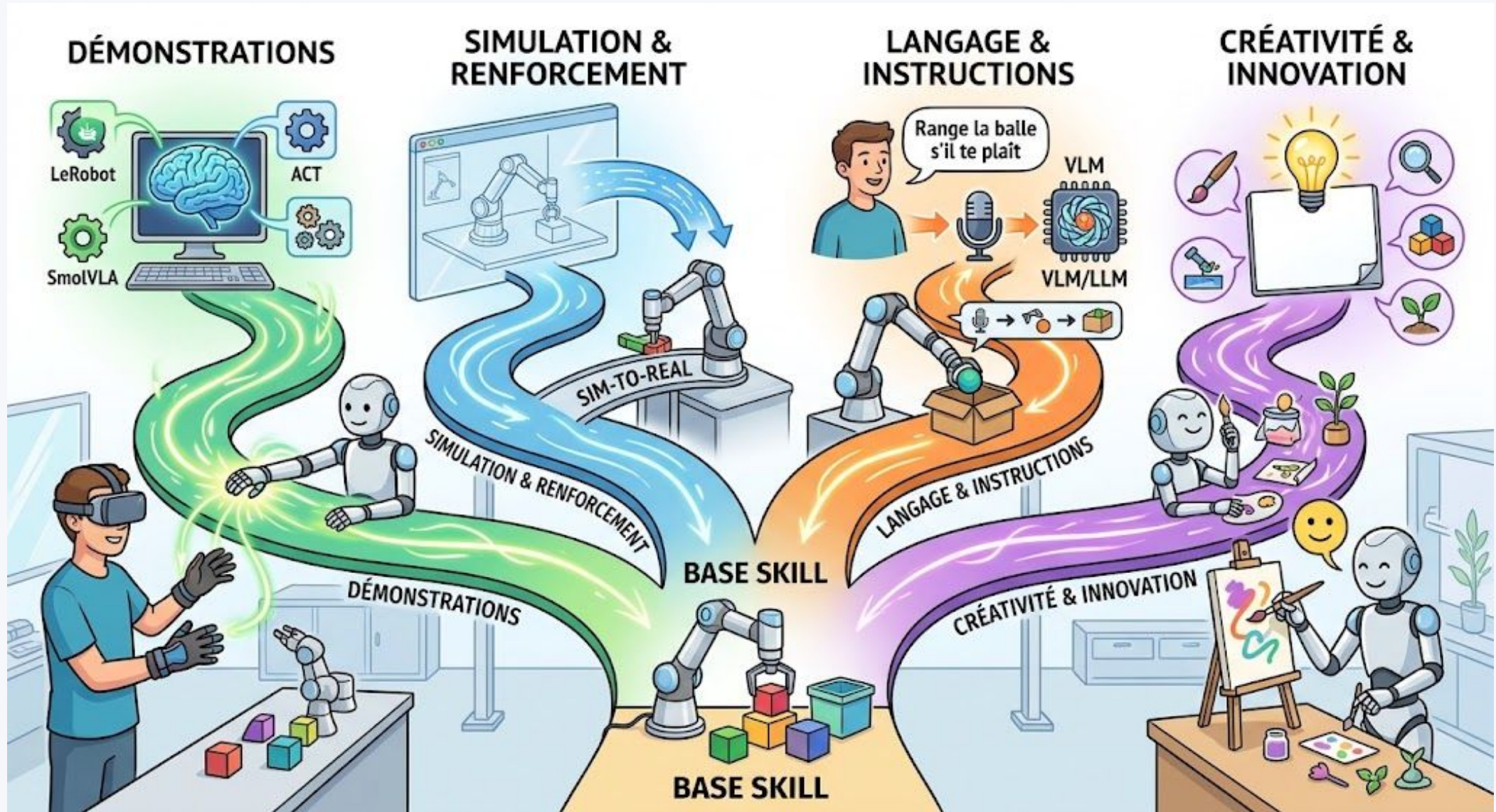
< Part 7 />

Going further

{ <TECH TOGETHER TOMORROW> }



< After pick & place />



20~30

minutes, in groups

Your idea must be

- Feasible in a few weeks
- Demonstrable on video
- A real robotics skill

Your constraints

1 arm

5 DoF + gripper

Low payload

small, light objects

Small workspace

a tabletop

1 camera

+ whatever you add

Your code

ROS 2, sim first

Safety

near humans: slow

1

The idea

in one sentence

2

The tech

what to add to the pipeline

3

The risk

what might not work



Achievable examples

Color sorting · Tic-tac-toe vs the robot · Candy handout · Drawing · Tidying a workbench

Tools

Phase 1 is your basis + IA in control or perception / Add language command / Path or Motion planning

***Be very ambitious ! Do something cool for you and your resume/cv !
If it does not work perfectly in the end, it's ok ;)***

{EPITECH}

<SINCE 1999/>

< Part 7 />

Going further

{ <TECH TOGETHER TOMORROW> }



< End-of-day check-in />

1

What works?

show it

2

What blocks?

3

Next steps

until online class #1

~5 minutes per group